

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Coding Challenge Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 1 – Coding Challenge Task
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
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Section / Batch:	2024	Date:	22-08-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Write clear code for the given problem.
- Analyze the Time complexity and Space complexity of your code using dynamic programming and greedy strategies

Question Type	Focus Area	Questions
Coding Challenge Task	Focus on Code and analyze optimization problems using dynamic programming and greedy strategies	Q1

SECTION A – Coding Challenge Task

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Coding Challenge Task

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%				
Algorithm	25%				
Code Implementation	20%				
Competitive Performance	20%				
Test Case Handling	15%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO4 AT1-CODE CHALLENGE TASK

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QUESTIONS

Q1

Rod Cutting Problem
5 marks

1/1 answered

Q1 Rod Cutting Problem

5 marks

A manufacturer has a rod of a fixed length and a list of prices corresponding to different possible piece lengths. The rod may be cut into smaller parts in any valid way, and the objective is to maximize the total selling price obtained from the cuts. The problem requires finding the most profitable combination of cuts. Use Dynamic Programming to determine the maximum obtainable value. Input Format: n = length of rod price[] = prices of pieces from length 1 to n Sample Input: n = 4 price = 1 5 8 9 Expected Output: Maximum Profit = 10

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] price = new int[n + 1];
8         for (int i = 1; i <= n; i++) {
9             price[i] = sc.nextInt();
10        }
11
12        int[] dp = new int[n + 1];
13        dp[0] = 0;
14
15        for (int len = 1; len <= n; len++) {
16            int maxVal = Integer.MIN_VALUE;
17            for (int cut = 1; cut <= len; cut++) {
18                maxVal = Math.max(maxVal, price[cut] + dp[len - cut]);
19            }
20            dp[len] = maxVal;
21        }
22
23        System.out.println("Maximum Profit = " + dp[n]);
24    }
25 }
```

[Submitted](#)

Input

4
1 5 8 9

Output

Maximum Profit = 10

Goal: Cut a rod of length n into pieces to maximize total selling price.

Approach — Bottom-up DP:

$dp[len]$ = best possible profit for a rod of length len .

Base case: $dp[0] = 0$ (no rod, no profit).

For every length from 1 to n , try every possible first cut (cut from 1 to len), and take the best of $price[cut] + dp[len - cut]$ — i.e., the value of that first piece plus the best profit from optimally cutting the remaining rod.

By the time you compute $dp[len]$, all smaller dp values are already known (that's what makes it DP instead of brute-force recursion).

Complexity: $O(n^2)$ time, $O(n)$ space — much better than the exponential brute-force of trying every cut combination.

Why it works: The problem has optimal substructure — the best way to cut a rod of length len is built from the best way to cut smaller rods.